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Machines	Units of power when on	Backup generator
A & B	15 + 18 = 33	On
B & C	18 + 12 = 30	Off
C & A	12 + 15 = 27	Off
A & B & C	15 + 18 + 12 = 45	On

Thus, the backup generator will be on whenever Machines A and B are both on, this being true regardless of whether Machine C is on. We are given that Machine A is on for 1 minute every 18 minutes and Machine B is on for 1 minute every 15 minutes. Therefore, if Machines A and B are both on for a certain minute, then the following are the minutes when these machines are again on.

Minutes when Machine A is on: 1st, 19th, 37th, 55th, 73rd, $\mathbf{91^{st}}$, 109th, …

Minutes when Machine B is on: 1st, 16th, 31st, 46th, 61st, 76th, $\mathbf{91^{st}}$, 106th, …

Therefore, the next time Machines A and B are both on is the 91st minute, which is 90 minutes after the first minute.

Alternatively, Machine A is on every $18 = (2)(3^2)$ minutes and Machine B is on every 15 = (3)(5) minutes, so the machines are both on every $(2)(3^2)(5) = 90$ minutes (least common multiple of 18 and 15).

The correct answer is C.

•

In a certain region, the number of children who have been vaccinated against rubella is twice the number who have been vaccinated against mumps. The number who have been vaccinated against both is twice the number who have been vaccinated against mumps. If 5,000 have been vaccinated against both, how many have been vaccinated against rubella only?

A. 2,500

B. 7,500

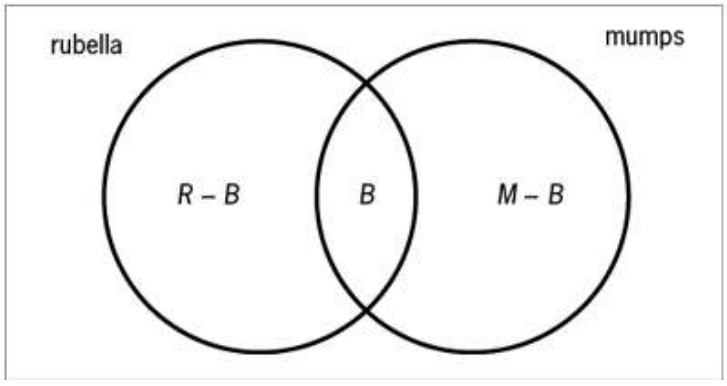
C. 10,000

D. 15,000

E. 17,500

Algebra **Simultaneous Equations; Sets (Venn Diagrams)**

Let R, M, and B be the number, respectively, of children who have been vaccinated against rubella, against mumps, and against both. Then the number of children who have been vaccinated against rubella only is $R - B$ and the number of children who have been vaccinated against mumps only is $M - B$, as shown in the Venn diagram below.



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From the given information, we have (i) $R = 2M$, (ii) $B = 2(M - B)$, and (iii) $B = 5{,}000$. Substituting the value of B given in (iii) into (ii) gives $5{,}000 = 2(M - 5{,}000)$, or $M = 7{,}500$. Using this value of M in (i) gives $R = 2(7{,}500) = 15{,}000$. Therefore, the number vaccinated against rubella only is $R - B = 15{,}000 - 5{,}000 = 10{,}000$.

The correct answer is C.

•

Three boxes of supplies have an average (arithmetic mean) weight of 7 kilograms and a median weight of 9 kilograms. What is the maximum possible weight, in kilograms, of the lightest box?

A. 1

B. 2

C. 3

D. 4

E. 5

Algebra Statistics

Let x, y, and z be the weights, in kilograms, of the boxes, where $x \leq y \leq z$. Since the average weight of the boxes is 7 kilograms and the median weight of the boxes is 9 kilograms, we have $x + y + z = 3(7) = 21$ and $y = 9$. Hence, $x + 9 + z = 21$, or $z = 12 - x$. Since $z = 12 - x$, $z \geq y$, and $y = 9$, it follows that $12 - x \geq 9$, or $x \leq 3$. Therefore, the maximum possible value of x, which is the maximum possible weight of the lightest box in kilograms, is less than or equal to 3. Since $x = 3$ is a possible value of x, which can be seen by considering $x = 3$, $y = 9$, and $z = 9$, the maximum possible weight of the lightest box is equal to 3 kilograms.

The correct answer is C.

Stock	Number of Shares
V	68
W	112
X	56
Y	94
Z	45

•

The table shows the number of shares of each of the 5 stocks owned by Mr. Sami. If Mr. Sami were to sell 20 shares of Stock X and buy 24 shares of Stock Y, what would be the increase in the range of the numbers of shares of the 5 stocks owned by Mr. Sami?

A. 4

B. 6

C. 9

D. 15

E. 20

Arithmetic Statistics

The table below shows the numbers of shares of the 5 stocks before and after the change in the numbers of shares of Stocks X and Y.

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Stocks X and Y.

Stock	Number of Shares (before)	Number of Shares (after)
V	68	68
W	112	112
X	56	36
Y	94	118
Z	45	45

Before the change in the number of shares of Stocks X and Y, the range was 112 − 45 = 67. After the change in the number of shares of Stocks X and Y, the range was 118 − 36 = 82. Therefore, the increase in the range is 82 − 67 = 15.

The correct answer is D.

•

Last year, sales at Company X were 10% greater in February than in January, 15% less in March than in February, 20% greater in April than in March, 10% less in May than in April, and 5% greater in June than in May. In which month were sales closest to the sales in January?

A. February

B. March

C. April

D. May

E. June

Arithmetic Percents; Estimation

Let N be the sales at Company X in January. The table gives numerical expressions for sales in each of the months from January through June. For example, since sales in March were 15% less than sales in February, sales in March can be obtained by multiplying sales in February by 0.85.

Month	Sales
January	N
February	$(1.1)N$
March	$(0.85)(1.1)N$
April	$(1.2)(0.85)(1.1)N$
May	$(0.9)(1.2)(0.85)(1.1)N$
June	$(1.05)(0.9)(1.2)(0.85)(1.1)N$

The month after January with sales closest to sales in January will be the month after January having sales closest to N. To this end, the following two results will be helpful.

a. (0.85)(1.1) = 0.935

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b. (0.85)(1.2) = 1.02

From (a) it follows that sales in January were closer to sales in March than to sales in February, since 1 is closer to 0.935 than to 1.1. Hence, the month with sales closest to January is NOT February. From (b) it follows that sales in January were closer to sales in February than to sales in April, since $(1.2)(0.85)(1.1) = (1.02)(1.1) > 1.1$.

Hence, the month with sales closest to January is NOT April. For May, because (0.9)(1.1) = (1 − 0.1)(1 + 0.1) = 1^2 − $(0.1)^2$ = 0.99, it follows from (b) that (0.9)(1.2)(0.85)(1.1) = (0.99)(1.02) = (0.99)(1 + 0.02), which is 0.99 plus a little less than 0.02 (0.99 × 0.02 is 99% of 0.02). Thus, sales in January were closer to sales in May (between N and $1.02N$) than to sales in March (sales were $0.935N$). Hence, the month with sales closest to January is NOT March. Finally, because sales in June were greater than sales in May, and both were greater than N, sales in January were closer to sales in May than to sales in June. Hence, the month with sales closest to January is NOT June. Therefore, the month with sales closest to January is May.

Alternatively, multiply the indicated products and determine which is closest to N.

Month	Sales
January	N
February	$1.1N$
March	$0.935N$
April	$1.122N$
May	$1.0098N$
June	$1.06029N$

The correct answer is D.

•

If s and t are integers greater than 1 and each is a factor of the integer n, which of the following must be a factor of n^{st}?

I. s^t

II. $(st)^2$

III. $s + t$

A. None

B. I only

C. II only

D. III only

E. I and II

Algebra Factoring; Simplifying Algebraic Expressions

Since each of s and t is a factor of n, it follows that there are positive integers a and b such that $n = as$ and $n = bt$.

I. MUST BE A FACTOR: nst divided by st is an integer, since $\frac{n^{st}}{s^t} = \frac{(as)^{st}}{s^t} = \frac{a^{st}s^{st}}{s^t} = a^{st}s^{st-t} = a^{st}s^{t(s-1)}$ and each of a^{st} and $st(s-1)$ is an integer. Note that $st(s-1)$ is an integer because s is greater than 1, and hence the exponent $t(s-1)$ is not negative.

II. MUST BE A FACTOR: Since $s \geq 2$ and $t \geq 2$, we have $st \geq 4$, or $st - 4 \geq 0$. Hence, n^{st-4} is an integer because the exponent $st - 4$ is not negative. Therefore, because $n^{st} = n^4 n^{st-4}$, it follows that n^4 is a factor of

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because the exponent $st - 4$ is not negative. Therefore, because $n^{st} = n^4 \cdot n^{st-4}$, it follows that n^4 is a factor of n^{st}. Also, because $n^4 = (n)(n)(n)(n) = (as)(as)(bt)(bt) = (ab)^2(st)^2$, it follows that $(st)^2$ is a factor of n^4. Finally, because $(st)^2$ is a factor of n^4 and n^4 is a factor of n^{st}, it follows that $(st)^2$ is a factor of n^{st}.

III. CAN FAIL TO BE A FACTOR. Let $s = 2$, $t = 3$, and $n = 6$. Then s and t are integers greater than 1 and each is a factor of n. However, $s + t = 5$ is not a factor of $n^{st} = 6^6$.

The correct answer is E.

•

How many 4-digit positive integers are there in which all 4 digits are even?

A. 625

B. 600

C. 500

D. 400

E. 256

Arithmetic Elementary Combinatorics

There are 4 possibilities for the left-most digit (2, 4, 6, and 8) and 5 possibilities for each of the 3 remaining digits (0, 2, 4, 6, and 8). Therefore, the number of 4-digit positive integers in which all 4 digits are even is 4 × 5 × 5 × 5 = 500.

The correct answer is C.

•

If $0 < r < 1 < s < 2$, which of the following must be less than 1?

I. $\frac{r}{s}$

II. rs

III. $s - r$

A. I only

B. II only

C. III only

D. I and II

E. I and III

Arithmetic Inequalities

I. MUST BE LESS THAN 1: We have $0 < r < s$ from the given information. Dividing throughout by the positive number s gives $0 < \frac{r}{s} < 1$.

II. CAN FAIL TO BE LESS THAN 1: If $r = 0.9$ and $s = 1.9$, then $0 < r < 1 < s < 2$ and $rs = 1.71$ is not less than 1.

III. CAN FAIL TO BE LESS THAN 1: If $r = 0.1$ and $s = 1.9$, then $0 < r < 1 < s < 2$ and $s - r = 1.8$ is not less than 1.

The correct answer is A.

•

Last month, 15 homes were sold in Town X. The average (arithmetic mean) sale price of the homes was $150,000 and the median sale price was $130,000. Which of the following statements must be true?

I. At least one of the homes was sold for more than $165,000.

II. At least one of the homes was sold for more than $130,000 and less than $150,000.

III. At least one of the homes was sold for less than $130,000.

A. I only

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A. I only

B. II only

C. III only

D. I and II

E. I and III

Arithmetic Statistics

I. MUST BE TRUE: Since the average sale price of the 15 homes was \$150,000, the total of the 15 sale prices was 15(\$150,000) = \$2,250,000. Also, since the median sale price of the 15 homes was \$130,000, 7 of the homes each had a sale price less than or equal to \$130,000, 7 of the homes each had a sale price greater than or equal to \$130,000, and the remaining home had a sale price equal to \$130,000.

$$\underbrace{\leq \$130{,}000 \text{ each}}_{7 \text{ homes}} \underbrace{= \$130{,}000}_{1 \text{ home}} \underbrace{\geq \$130{,}000 \text{ each}}_{7 \text{ homes}}$$

If none of the homes was sold for more than \$165,000, then the total of the 15 sale prices would be less than or equal to 7(\$130,000) + \$130,000 + 7(\$165,000) = \$2,195,000, which is less than \$2,250,000, contradicting what we know is the total of the 15 sale prices. Therefore, it is not possible for none of the homes to be sold for more than \$165,000, and thus it must be true that at least one of the homes was sold for more than \$165,000.

II. CAN BE FALSE: If 13 of the sale prices were each \$130,000 and 2 of the sale prices were each \$280,000, then the median of the sale prices would be \$130,000 and the average of the sale prices would be $\frac{13(\$130{,}000) + 2(\$280{,}000)}{15} = \$150{,}000.$ However, in this example none of the homes was sold for more than \$130,000 and less than \$150,000.

III. CAN BE FALSE: If 13 of the sale prices were each \$130,000 and the 2 remaining sale prices were \$140,000 and \$420,000, then the median of the sale prices would be \$130,000 and the average of the sale prices would be $\frac{13(\$130{,}000) + \$140{,}000 + \$420{,}000}{15} = \$150{,}000.$

However, in this example none of the homes was sold for less than \$130,000.

The correct answer is A.

•

Pumps A, B, and C operate at their respective constant rates. Pumps A and B, operating simultaneously, can fill a certain tank in $\frac{6}{5}$ hours; Pumps A and C, operating simultaneously, can fill the tank in $\frac{3}{2}$ hours; and Pumps B and C, operating simultaneously, can fill the tank in 2 hours. How many hours does it take Pumps A, B, and C, operating simultaneously, to fill the tank?

A. $\frac{1}{3}$

B. $\frac{1}{2}$

C. $\frac{2}{3}$

D. $\frac{5}{6}$

E. 1

Algebra Work Problem

Let t_A, t_B, and t_C be the number of hours, respectively, for Pumps A, B, and C operating alone to fill the tank, and let t be the number of hours for Pumps A, B, and C operating simultaneously to fill the tank.

1 1 1 5

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Since Pumps A and B, operating simultaneously, can fill the tank in $\frac{6}{5}$ hours, it follows that $\frac{1}{t_A} + \frac{1}{t_B} = \frac{1}{\frac{6}{5}} = \frac{5}{6}$, and similarly for the other two pairs of pumps we are given information about.

$$\frac{1}{t_A} + \frac{1}{t_B} = \frac{5}{6}$$
$$\frac{1}{t_A} + \frac{1}{t_C} = \frac{2}{3}$$
$$\frac{1}{t_B} + \frac{1}{t_C} = \frac{1}{2}$$

Adding these 3 equations gives $2\left(\frac{1}{t_A} + \frac{1}{t_B} + \frac{1}{t_C}\right) = \frac{5}{6} + \frac{2}{3} + \frac{1}{2} = 2$, or $\frac{1}{t_A} + \frac{1}{t_B} + \frac{1}{t_C} = 1$. Since $\frac{1}{t_A} + \frac{1}{t_B} + \frac{1}{t_C} = \frac{1}{t}$, it follows that $\frac{1}{t} = 1$, or $t = 1$.

The correct answer is E.

•

If n is a positive integer, then $(-2^n)^{-2} + (2^{-n})^2 =$

A. 0

B. 2^{-2n}

C. 2^{2n}

D. 2^{-2n+1}

E. 2^{2n+1}

Algebra Simplifying Algebraic Expressions

$(-2^n)^{-2} + (2^{-n})^2$	given
$= \frac{1}{(-2^n)^2} + \left(\frac{1}{2^n}\right)^2$	use $a^{-b} = \frac{1}{a^b}$
$= \frac{1}{(-1)^2(2^n)^2} + \left(\frac{1}{2^n}\right)^2$	$-2^n = (-1)(2^n)$ and use $(ab)^c = a^c b^c$
$= \frac{1}{(2^n)^2} + \frac{1}{(2^n)^2}$	$(-1)^2 = 1$ and use $\left(\frac{a}{b}\right)^c = \frac{a^c}{b^c}$
$= \frac{2}{(2^n)^2}$	adding fractions with a common denominator
$= \frac{2}{2^{2n}}$	use $(a^b)^c = a^{bc}$
$= 2^{1-2n}$	$2 = 2^1$ and use $\frac{a^b}{a^c} = a^{b-c}$

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$$= 2^{-2n+1} \qquad 1 - 2n = -2n + 1$$

The correct answer is D.

•

Which of the following is equal to $\left(\frac{2\sqrt[3]{3}}{\sqrt{2}}\right)^3$?

A. $\frac{3\sqrt{2}}{2}$

B. $3\sqrt{2}$

C. $6\sqrt{2}$

D. 12

E. $12\sqrt{2}$

Arithmetic Operations on Radical Expressions

$$\left(\frac{2\sqrt[3]{3}}{\sqrt{2}}\right)^3 = \frac{(2\sqrt[3]{3})^3}{(\sqrt{2})^3} \qquad \text{use } \left(\frac{a}{b}\right)^c = \frac{a^c}{b^c}$$

$$= \frac{(2)^3(\sqrt[3]{3})^3}{(\sqrt{2})(\sqrt{2})(\sqrt{2})} \qquad \text{use } (ab)^c = a^c b^c$$

$$= \frac{(8)(3)}{2\sqrt{2}} \qquad (\sqrt{2})(\sqrt{2}) = 2 \text{ and } (\sqrt[3]{3})^3 = 3$$

$$= \frac{12}{\sqrt{2}} \qquad \text{reduce}$$

$$= \frac{12}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} \qquad \text{multiply by 1}$$

$$= \frac{12\sqrt{2}}{2} \qquad (\sqrt{2})(\sqrt{2}) = 2$$

$$= 6\sqrt{2} \qquad \text{reduce}$$

The correct answer is C.

•

Dara ran on a treadmill that had a readout indicating the time remaining in her exercise session. When the readout indicated 24 min 18 sec, she had completed 10% of her exercise session. The readout indicated which of the following when she had completed 40% of her exercise session?

A. 10 min 48 sec

B. 14 min 52 sec

C. 14 min 58 sec

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D. 16 min 6 sec

E. 16 min 12 sec

Arithmetic Rate Problem

Since 24 min 18 sec (or 1,458 sec) represents 90% of the total time for the exercise session, it follows that the total time for the exercise session was $\frac{1,458}{0.9}$ sec = 1,620 sec.

Thus, when she had completed 40% of the exercise session, the readout would have indicated 60% of the total time, which is 0.6(1,620 sec) = 972 sec = 16 min 12 sec.

The correct answer is E.

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